THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which ratio is greater: 9:11 or 13:16? 1

(a) cannot be compared **(b)** 13:16
(c) both are equal **(d)** 9:11

2. If A:B = 5:6 and B:C = 2:3, then A:B:C is: 1

(a) 5:6:9 **(b)** 5:8:3 **(c)** 10:18:9 **(d)** 5:6:3

3. 28 men can do a piece of work in 6 days. 21 men will do it in (days): 1

(a) 8 **(b)** 11 **(c)** 168 **(d)** 7

4. On a map the scale is 1 cm = 45 km. Two places 360 km apart are shown on the map at a distance (in cm) of: 1

(a) 315 **(b)** 9 **(c)** 16200 **(d)** 8

5. In a survey of 72 students, 11 chose cricket. In a pie chart, the central angle of the cricket sector (in degrees) is: 1

(a) 33 **(b)** 110 **(c)** 55 **(d)** 65

6. ₹ 4500 is divided among A, B and C in the ratio 2:3:10. B's share (in ₹) is: 1

(a) 1500 **(b)** 3000 **(c)** 600 **(d)** 900

7. A car moving at a constant speed covers 325 km in 5 hours. In 7 hours it will cover (km): 1

(a) 332 **(b)** 465 **(c)** 455 **(d)** 2275

8. Which ratio is greater: 7:10 or 11:15? 1

(a) 11:15 **(b)** cannot be compared
(c) both are equal **(d)** 7:10

9. If A:B = 2:5 and B:C = 3:7, then A:B:C is: 1

(a) 2:8:7 **(b)** 2:5:7 **(c)** 6:15:35 **(d)** 6:35:12

10. 25 men can do a piece of work in 12 days. 30 men will do it in (days): 1

(a) 10 **(b)** 300 **(c)** 13 **(d)** 24

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. y is directly proportional to x. If $y = 63$ when $x = 9$, find the constant of proportionality and y when $x = 14$. 2

12. Which ratio is greater: 4:5 or 13:17? 2

13. x and y are inversely proportional. When $x = 12$, $y = 25$. Find y when $x = 10$. 2

14. y is directly proportional to x. If $y = 55$ when $x = 5$, find the constant of proportionality and y when $x = 8$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. On a map of scale 1 cm = 6 km, a rectangular park measures 5 cm by 3 cm. Find the actual length, breadth and perimeter of the park. 3

16. A family's monthly expenditure: food 45%, rent 20%, education 5%, others 30%. Find the central angle of each sector of a pie chart. 3

17. A school bus travelling at 24 km/h takes 35 minutes to reach the school. At what speed must it travel to reach in 28 minutes? 3

OR

On a map of scale 1 cm = 10 km, a rectangular park measures 12 cm by 7 cm. Find the actual length, breadth and perimeter of the park. 3

18. A family's monthly expenditure: food 20%, rent 40%, education 15%, others 25%. Find the central angle of each sector of a pie chart. 3

OR

A school bus travelling at 50 km/h takes 27 minutes to reach the school. At what speed must it travel to reach in 18 minutes? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. ₹ 1040 is divided among A, B and C such that A:B = 3:4 and B:C = 2:3. Find the share of each. 5

OR

The marks of a student are: Walk 9, Cycle 12, Bus 15, Car 6, Train 3. Total 45. Find the central angle of every sector of a pie chart and the percentage of the total taken by Walk. 5

20. ₹ 2820 is divided among A, B and C such that A:B = 4:5 and B:C = 3:4. Find the share of each. 5

OR

The marks of a student are: Red 18, Blue 24, Green 12, Pink 10, White 8. Total 72. Find the central angle of every sector of a pie chart and the percentage of the total taken by Red. 5

END OF QUESTION PAPER — Total: 40 Marks

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